## Type MS · Solving log and exponential equations with your graph

### Answer 1 of 12 · Question 5 part c · 3 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

| | | |
|---|---|---|
| 5 (a) | <table><tr><td>$x$</td><td>0</td><td>0.25</td><td>0.5</td><td>0.75</td><td>1</td></tr><tr><td>$y$</td><td>0.14</td><td>**0.29**</td><td>**0.61**</td><td>**1.28**</td><td>2.72</td></tr></table> | first B1<br>2 values correct<br><br>second B1<br>all 3 values correct<br>[2] |
| (b) | Each point plotted correctly within the correct small square<br><br>Smooth curve through the points | B1ft<br><br>B1ft<br>[2] |
| (c) | $e^{3x-2} = 3 - x$<br><br>$y = 3 - x$<br><br>0.9 | M1<br><br>M1<br><br>A1ft<br>[3] |
| | | **Total 7 marks** |

## Type MS · Solving log and exponential equations with your graph

### Answer 1 of 12 · Question 5 part c · 3 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

Type MS · Solving log and exponential equations with your graph

Answer 2 of 12 · Question 8 part d · 4 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part d in this mark scheme extract

| Number | | |
|---|---|---|
| 8 (a) | <table><tr><td>$x$</td><td>0</td><td>0.25</td><td>0.5</td><td>1</td><td>1.5</td><td>2</td><td>3</td></tr><tr><td>$y$</td><td>2</td><td>**2.41**</td><td>**2.69**</td><td>3.10</td><td>3.39</td><td>3.61</td><td>**3.95**</td></tr></table> | B2<br>[2] |
| (b) | Points plotted within half of a square<br>Points joined together in a smooth curve | B1ft<br>B1ft<br>[2] |
| (c) | $\ln(2x+1) = 3x-4 \Rightarrow \ln(2x+1)+2 = 3x-2$<br>Graph of $y = 3x-2$ drawn. Intersection point is at $x = 1.8$ or $1.9$ (Accept either) | M1<br>M1A1<br>[3] |
| (d) | $e^{(6-x)} = (2x+1)^2 \Rightarrow 6-x = \ln(2x+1)^2 \Rightarrow 6-x = 2\ln(2x+1)$<br>$\Rightarrow \ln(2x+1)+2 = 5-\frac{x}{2}$<br>Graph of $y = 5-\frac{x}{2}$ drawn. Intersection point is at $x = 2.4$ or $2.5$ (Accept either) | M1<br>M1<br>M1A1<br>[4] |
| | | Total 11 marks |

## Type MS · Solving log and exponential equations with your graph

### Answer 2 of 12 · Question 8 part d · 4 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part d in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

Type MS · Solving log and exponential equations with your graph

Answer 3 of 12 · Question 7 part c · 6 marks

2023 JAN · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

| | | |
|---|---|---|
| 7 a | <table><tr><td>-6</td><td>-5</td><td>-4</td><td>–3</td><td>–2</td><td>–1</td><td>0</td></tr><tr><td>4</td><td>3.59</td><td>3.26</td><td>**3**</td><td>**2.79**</td><td>**2.63**</td><td>2.5</td></tr></table> | B2 (2) |
| b | Points plotted within a half square<br>Joined with a smooth curve | B1 ft<br>B1 ft<br>(2) |
| c | $\left(\log_2(2x+2)^3 + x + 3 = 0\right)$ | |
| | $3\log_2(2x+2) = -x-3$ or $\log_2(2x+2) = \frac{-x}{3} - 1$ | M1 |
| | $\frac{\log_{0.5}(2x+2)}{\log_{0.5}2} = -\frac{x}{3} - 1$ or $2x+2 = 2^{-\frac{x}{3}-1}$ or $\frac{1}{2x+2} = 2^{\frac{x}{3}+1}$ | M1 |
| | $\left(\log_{0.5}(2x+2) = \frac{x}{3}+1 \Rightarrow\right) 2x+2 = 0.5^{\frac{x}{3}+1}$ | M1 |
| | $2x+4 = 0.5^{\left(\frac{x}{3}+1\right)} + 2$ | A1 |
| | $y = 2x+4$ drawn | M1 |
| | Intersect at $x = -0.8 / -0.7$ | A1<br>(6) |
| ALT | $3\log_2(2x+2) = -x-3$ or $\log_2(2x+2) = \frac{-x}{3} - 1$ | M1 |
| | $\frac{\log_{0.5}(2x+2)}{\log_{0.5}2} = -\frac{x}{3} - 1 \left(\Rightarrow \log_{0.5}(2x+2) = \frac{x}{3}+1\right)$ | M1 |
| | $\left(y-2 = 0.5^{\left(\frac{x}{3}+1\right)} \Rightarrow\right) \log_{0.5}(y-2) = \frac{x}{3}+1 \Rightarrow \log_{0.5}(y-2) = \log_{0.5}(2x+2)$ | M1 |
| | $y-2 = 2x+2$ | A1 |
| | As main scheme | M1 A1 |
| | | **Total 10 marks** |

## Type MS · Solving log and exponential equations with your graph

### Answer 3 of 12 · Question 7 part c · 6 marks

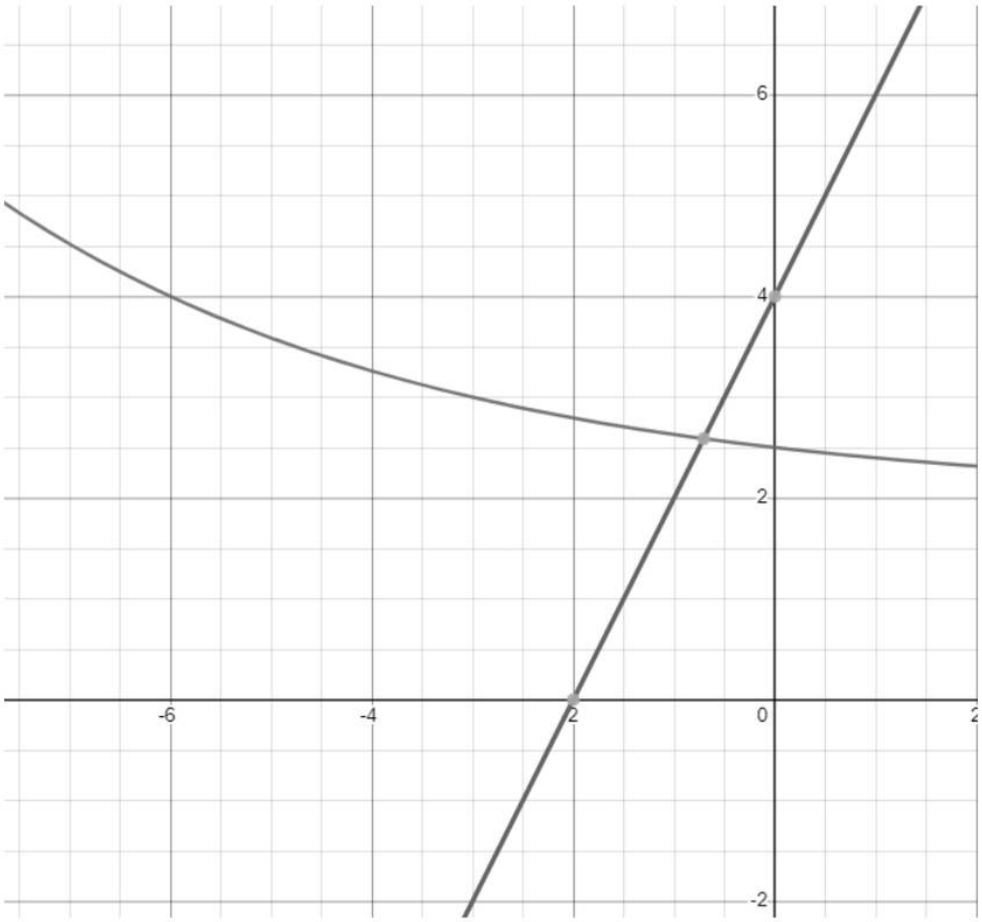
2023 JAN · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | B2 | For all 3 values in the table correct to 2 d.p. Allow 3.0 or 3.00 for "3"<br>(B1 for 2 values) |
| (b) | B1 ft | For all of the points plotted within half a square, allow use of their values.<br>Points must be checked carefully, including using the zoom tool on ePen if necessary. |
| | B1 ft | For all of their points joined with a smooth curve. The curve must pass through each of their points to within half a square.<br>Be cautious to not award this mark if straight lines are drawn between the points plotted. |
| (c) | M1 | For use of $\log_a x^k = k\log_a x$ and an attempt to rearrange.<br>Accept an equation of the form<br>$\pm 3\log_2(2x+2) = \pm x \pm 3$ or $\pm\log_2(2x+2) = \frac{\pm x}{3} \pm 1$ |
| | M1 | For correctly dividing by 3 (if not already done) and correct use of<br>$\log_b a = \frac{\log_{0.5}(a)}{\log_{0.5}(b)}$ with their equation to get an expression of the form<br>$\frac{\log_{0.5}(2x+2)}{\log_{0.5}2} = \pm\frac{x}{3} \pm 1$ or<br>For correctly dividing by 3 (if not done) and correctly converting their equation to exponential form get an equation of the form $2x+2 = 2^{\pm\frac{x}{3}\pm 1}$ or $\frac{1}{2x+2} = 2^{\pm\frac{x}{3}\pm 1}$ |
| | M1 | For reaching an equation of the form $2x+2 = 0.5^{\pm\frac{x}{3}\pm 1}$ |
| | A1 | For $2x+4 = 0.5^{\left(\frac{x}{3}+1\right)} + 2$ |
| | M1 | For $y = 2x+4$ drawn. Correct line drawn can imply any of the 4 previous marks.<br>Allow any line of $y = 2x + c$ to gain this mark.<br>**If the candidate draws any line where $c$ is other than 4, they must have gained the previous 3 method marks.** |
| | A1 | For $x = -0.8 / -0.7$ |
| ALT | M1 | For use of $\log_a x^k = k\log_a x$ and an attempt to rearrange.<br>Accept an equation of the form<br>$\pm 3\log_2(2x+2) = \pm x \pm 3$ or $\pm\log_2(2x+2) = \frac{\pm x}{3} \pm 1$ |
| | M1 | For correctly dividing by 3 (if not already done) and correct use of<br>$\log_b a = \frac{\log_{0.5}(a)}{\log_{0.5}(b)}$ with their equation to get an expression of the form<br>$\frac{\log_{0.5}(2x+2)}{\log_{0.5}2} = \pm\frac{x}{3} \pm 1$ |
| | M1 | For an attempt to rearrange and converting to log form to reach<br>$\log_{0.5}(y \pm 2) = \pm\frac{x}{3} \pm 1 \Rightarrow \log_{0.5}(y \pm 2) = \pm\log_{0.5}(2x+2)$ |
| | A1 | For the correct equation show. |
| | M1<br>A1 | As main scheme. |

## Type MS · Solving log and exponential equations with your graph

### Answer 3 of 12 · Question 7 part c · 6 marks

2023 JAN · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract



## Type MS · Solving log and exponential equations with your graph

Answer 3 of 12 · Question 7 part c · 6 marks

2023 JAN · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| | | |

Type MS · Solving log and exponential equations with your graph

Answer 4 of 12 · Question 7 part c · 6 marks

2025 JUN · Paper 02 · extract 1 of 4 · find part c in this mark scheme extract

| | | |
|---|---|---|
| 7 (a) | $x$: 0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5<br>$y$: 4.26, **4.14**, 4, **3.83**, 3.63, **3.37**, 3, 2.37 | B2<br>(2) |
| (b) | Points plotted within half a square.<br>Points joined with a smooth curve. | B1ft<br>B1ft<br>(2) |
| (c) | $3^{2x-5} = (4-x)^3 \Rightarrow 2x-5 = 3\log_3(4-x)$ | M1M1 |
| | $\frac{2x-5}{3} + 3 = \log_3(4-x) + 3 \Rightarrow \frac{2x+4}{3} = \log_3(4-x) + 3$ | M1A1 |
| | **ALT** | |
| | $\log_3(4-x) + 3 = mx + c$ | |
| | $\Rightarrow \log_3(4-x) = mx + (c-3)$ | |
| | $\Rightarrow (4-x) = 3^{mx+(c-3)}$ | [M1 |
| | $\Rightarrow (4-x)^3 = 3^{(mx+(c-3))3} = 3^{3mx+3c-9}$ | M1 |
| | $\Rightarrow 2x-5 = 3mx+3c-9 \Rightarrow 3m = 2$ and $-5 = 3c-9$ | M1 |
| | $\Rightarrow m = \frac{2}{3} \quad c = \frac{4}{3}$ | A1] |
| | Graph of $y = \frac{2x+4}{3}$ drawn. Intersection point is $x = 2.8$ | M1A1<br>(6)<br>**[10]** |

## Type MS · Solving log and exponential equations with your graph

### Answer 4 of 12 · Question 7 part c · 6 marks

2025 JUN · Paper 02 · extract 2 of 4 · find part c in this mark scheme extract

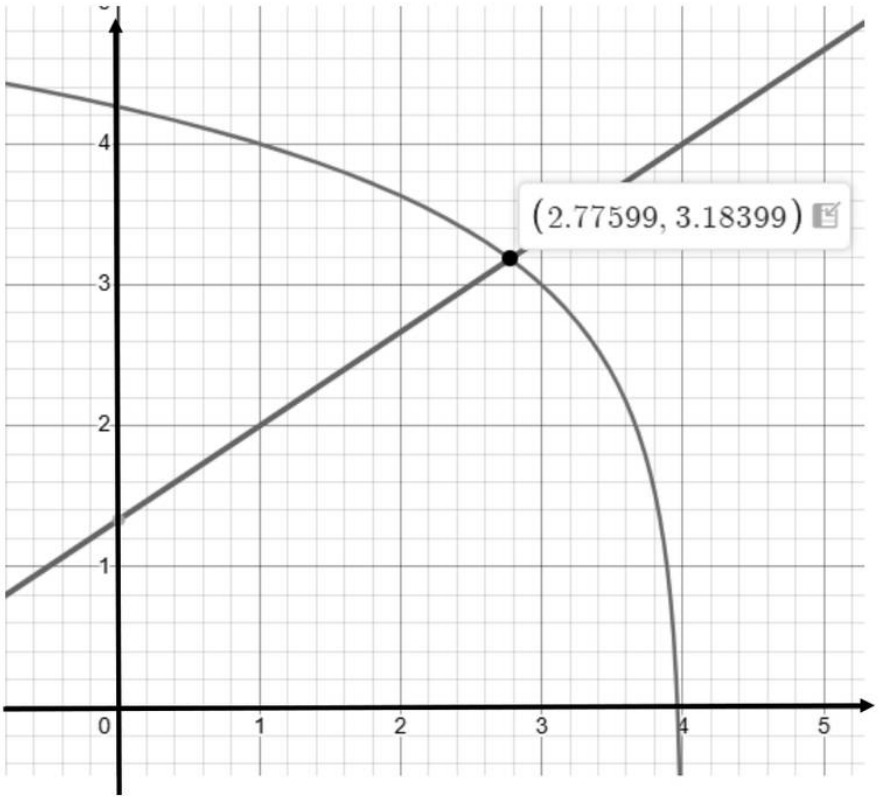
| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | B2 | All three points correct<br>(B1 for 2 points correct) |
| (b) | B1ft | Points plotted ft their table. Allow half a square tolerance. |
| | B1ft | Points joined together with a smooth curve. Allow half a square tolerance on joining up their points.<br>FT their table |
| (c) | M1 | For taking logs base 3 of both sides<br>$2x-5=\log_3(4-x)^3$ |
| | M1 | For applying the power law<br>$2x-5=3\log_3(4-x)$ |
| | ddM1 | For attempting to rearrange to the required form<br>An attempt is defined as dividing through by 3 and adding 3 to both sides. |
| | A1 | For the correct line, in any form<br>$\frac{2x-5}{3}+3=\log_3(4-x)+3\Rightarrow\left[(y=)\frac{2}{3}x+\frac{4}{3}\right]$ |
| | **ALT for first 4 marks** | |
| | M1 | Sets up the equation to obtain $\log_3(4-x)+3=mx+c$<br>And removes the log to obtain $(4-x)=3^{mx+(c-3)}$ |
| | M1 | Raises **both** sides to a power of 3.<br>Allow one error in this on the LHS |
| | M1 | Compares coefficients to obtain an equation in $m$ and an equation in $c$ |
| | A1 | Obtains the values $m=\frac{2}{3}$ $c=\frac{4}{3}$ |
| | **Draws the line** | |
| | M1 | For their line drawn provided it is of the form $y=kx\pm\frac{4}{3}$ for $k\neq 0,\ 1$ |
| | A1 | $x=2.8$ or 2.7<br>Calculator value is 2.776 |

## Type MS · Solving log and exponential equations with your graph

### Answer 4 of 12 · Question 7 part c · 6 marks

2025 JUN · Paper 02 · extract 3 of 4 · find part c in this mark scheme extract

**USEFUL GRAPH**



## Type MS · Solving log and exponential equations with your graph

### Answer 4 of 12 · Question 7 part c · 6 marks

2025 JUN · Paper 02 · extract 4 of 4 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

Type MS · Solving log and exponential equations with your graph

Answer 5 of 12 · Question 4 part c · 5 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

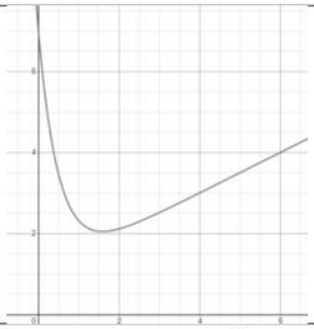
**4(a)**

| $x$ | 0 | 1 | 1.5 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|---|
| $y$ | 7 | **2.3** | 2.0 | **2.1** | **2.5** | 3 | **3.5** | 4 |

B2

[2]

**(b)**



B2ft

[2]

**(c)**

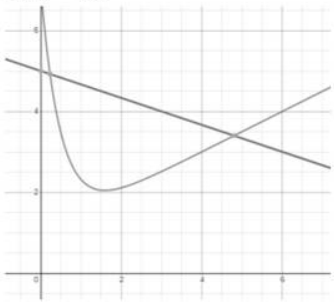
$$2x+\ln(24-5x)-\ln 36=0 \Rightarrow \ln\left(\frac{24-5x}{36}\right)=-2x$$

$$\Rightarrow \frac{24-5x}{36}=e^{-2x} \Rightarrow \frac{2}{3}-\frac{5x}{36}=e^{-2x}$$

$$\Rightarrow 4-\frac{5x}{6}=6e^{-2x}$$

$$\Rightarrow 5-\frac{x}{3}=\frac{x}{2}+6e^{-2x}+1 \Rightarrow y=5-\frac{x}{3}$$

$$\Rightarrow x=0.2,\quad 4.8$$



M1

M1A1

M1A1FT

[5]

**Total 9 marks**

## Type MS · Solving log and exponential equations with your graph

### Answer 5 of 12 · Question 4 part c · 5 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

| Part | Mark | Notes |
|---|---|---|
| (a) | B1 | At least two correct values rounded correctly. |
| | B1 | All values correct, rounded correctly. |
| (b) | B1ft | Their points plotted within half of one square. |
| | B1ft | Their points joined up in a smooth curve.<br>Note: do not allow ruled sections. |
| (c) | M1 | Combines the logs correctly and raises both sides to the power of base e |
| | M1 | Rearranges the equation to obtain the equation of the curve on one side and the equation of the straight line on the other. |
| | A1 | Achieves a correct equation of the straight line.<br>Note: equivalent forms are acceptable. |
| | M1 | Draws their line provided it is of the form $y = k \pm \frac{x}{3}$ when simplified. |
| | A1FT | Provided M1M1A1M1 awarded.<br>For both correct values of $x$ to 1 decimal place, their values **must** follow through from their graph.<br>0.2, 0.3, 0.4 and 4.8<br>Condone coordinates.<br>Note: correct values seen with no working scores M0M0A0M0A0 |

## Type MS · Solving log and exponential equations with your graph

Answer 5 of 12 · Question 4 part c · 5 marks

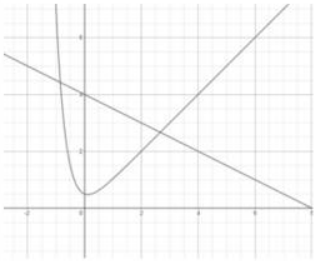
2023 JUN · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|

Type MS · Solving log and exponential equations with your graph

Answer 6 of 12 · Question 6 part whole · 7 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part whole in this mark scheme extract

| | | |
|---|---|---|
| 6 | $\log_2(8-3x)-1=-4x-1 \Rightarrow \log_2(8-3x)-\log_2 2=-4x-1$<br>$\Rightarrow \log_2\left(\frac{8-3x}{2}\right)=-4x-1$<br>$\Rightarrow \frac{8-3x}{2}=2^{-4x-1}$<br>$\Rightarrow 4-\frac{3}{2}x+x=x+2^{-(4x+1)}$<br>$\Rightarrow 4-\frac{1}{2}x=x+2^{-(4x+1)}$<br>Draws the line $y=4-\frac{x}{2}$<br>For the values $x=-0.8/0.9$ and $x=2.6/2.7$<br>  | M1<br>M1<br>M1<br>M1A1<br>M1<br>A1<br>[7] |
| | | **Total 7 marks** |

| Mark | Notes |
|---|---|
| | **General principles of marking this question for the first 5 marks**<br>**Award the first two marks when you see them.** |
| **M1** | Award the first M mark for dealing with the log wherever you see it<br>For example: $\log_2(8-3x)=-4x \Rightarrow 8-3x=2^{-4x}$ |
| **M1** | Award the second M mark for either adding – 1 to **both** sides and converting $-1=\log_2 2$ ,<br>**or** by dividing throughout by 2 and dealing with the power of – 1.<br>For example: $\frac{8-3x}{2}=\frac{2^{-4x}}{2} \Rightarrow 4-\frac{3x}{2}=2^{-4x-1}$ |
| **M1** | Award the third M mark for adding $x$ to **both** sides |
| **M1** | For simplifying and obtaining $(y)=x+2^{-4x-1}$ or $(y)=x+2^{-(4x+1)}$ on one side and a linear equation of the form $y=K\pm\frac{1}{2}x$ (may be unsimplified) where $K$ is a constant, on the other side. |
| **A1** | For obtaining the correct equation $y=4-\frac{1}{2}x$ (may be unsimplified) |
| **M1** | For drawing their line correctly<br>• Their line is in the form $y=k-\frac{x}{2}$ , $k$ nonzero constant (may be unsimplified)<br>• There must be at least one intersection point with the given curve. |
| **A1** | Correct $x$ values (both rounded to 1dp), ignore if $y$=…is present, accept also given as coordinates, even if their $y$ coordinates are incorrect. |

## Type MS · Solving log and exponential equations with your graph

Answer 6 of 12 · Question 6 part whole · 7 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part whole in this mark scheme extract

**ALT I** for first 5 marks

**M1:** Correctly removes log, achieves $2^{-4x}=8-3x$

**M1:** For dividing by 2 throughout $\frac{2^{-4x}}{2}=\frac{8-3x}{2} \Rightarrow 2^{-(4x+1)}=4-\frac{3}{2}x$

**M1:** For adding $x$ to both sides

**M1:** Rearranges leading to $x+2^{-4x-1}=4-\frac{x}{2}$

**A1**: For obtaining the correct equation $y=4-\frac{1}{2}x$ (may be unsimplified)

**ALT II** for first 5 marks

**M1**: Correctly removes log, achieves $2^{-4x}=8-3x$

**M1**: Rearrange the given curve to $y-x=2^{-4x}\times 2^{-1}$

**M1**: Attempts to make $2^{-4x}$ the subject from $y-x=2^{-4x}\times 2^{-1} \Rightarrow 2^{-4x}=\ldots$

**M1**: Equates the two resulting equations, then attempts to make y=…

**A1**: For obtaining the correct equation $y=4-\frac{1}{2}x$ (may be unsimplified)

For reference the two correct equations are: $\begin{cases} 2^{-4x}=8-3x \\ 2^{-4x}=2y-2x \end{cases} \Rightarrow 8-3x=2y-2x \Rightarrow y=4-\frac{x}{2}$

**ALT III** for first 5 marks

**M1:** Correctly removes log, achieves $2^{-4x}=8-3x$

**M1:** For $2^{-(4x+1)}=2^{-4x}\times 2^{-1}$ (may be embedded)

**M1:** Replaces $2^{-4x}$ with their $8-3x$

**M1:** Rearranges leading to y=…

**A1**: For obtaining the correct equation $y=4-\frac{1}{2}x$ (may be unsimplified)

## Type MS · Solving log and exponential equations with your graph

Answer 6 of 12 · Question 6 part whole · 7 marks

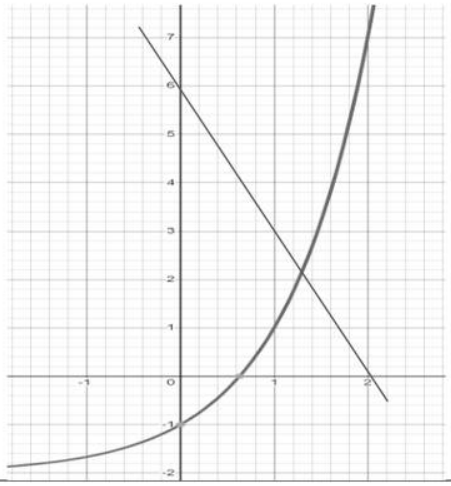
2024 JUN · Paper 02 (regional R) · extract 3 of 3 · find part whole in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|

Type MS · Solving log and exponential equations with your graph

Answer 7 of 12 · Question 8 part d · 5 marks

2025 NOV · Paper 01 · extract 1 of 3 · find part d in this mark scheme extract

| | | |
|---|---|---|
| **8(a)** | <table><tr><td>$x$</td><td>−1</td><td>−0.5</td><td>0</td><td>0.25</td><td>0.5</td><td>0.75</td><td>1</td><td>1.5</td><td>2</td></tr><tr><td>$y$</td><td>−1.7</td><td>−1.4</td><td>−1</td><td>−0.7</td><td>−0.3</td><td>0.3</td><td>1</td><td>3.2</td><td>7</td></tr></table> | B2<br>[2] |
| **(b)** |  | B2<br>[2] |
| **(c)** | $\log_3 4.5$<br>$y = 3^x - 2 \Rightarrow 3^x = y + 2 \Rightarrow x = \log_3(y+2)$<br>$\log_3 4.5 = \log_3(2.5+2) \Rightarrow y = 2.5$<br>$\Rightarrow x = 1.4$ | M1<br>M1<br>A1<br>[3] |
| **(d)** | $3^x - 2 = Ax + B \Rightarrow 3^x = Ax + (B+2) \Rightarrow x = \log_3(Ax + (B+2))$<br>$\Rightarrow \log_3(Ax + (B+2)) - x = 0$<br>$\log_3(8-3x) - x = 0$<br>$\Rightarrow A = -3 \quad 8 = B + 2 \Rightarrow B = 6$<br>$\Rightarrow y = 6 - 3x$<br><br>**ALT**<br><br>$\log_3(8-3x) - x = 0 \Rightarrow \log_3(8-3x) = x$<br>Taking exponents<br>$8 - 3x = 3^x \Rightarrow 8 - 3x - 2 = 3^x - 2 = y$<br>$6 - 3x = y$<br><br>Draws the line $y = 6 - 3x$<br>$x = 1.3$ | M1<br><br>M1<br><br>A1<br><br>[M1]<br><br>[M1]<br>[A1]<br><br>M1<br>A1<br>[5] |
| | | **Total 12 marks** |

## Type MS · Solving log and exponential equations with your graph

Answer 7 of 12 · Question 8 part d · 5 marks

2025 NOV · Paper 01 · extract 2 of 3 · find part d in this mark scheme extract

| Part | Mark | Notes |
|---|---|---|
| 8 (a) | B2 | B2 for all correct values<br>B1 for a minimum of two values |
| (b) | B1ft | All points plotted correctly within half of one square. |
| | B1ft | Their points joined by a smooth curve. |
| (c) | M1 | **In this part, there must be evidence of algebraic work, together with use of the graph.**<br>For using log laws to rearrange the given expression to find $x = \log_3(y+2)$ |
| | M1 | For deducing that the required value of $y$ is 2.5 and using their graph to find the value of $x$ at that point. |
| | A1 | For $x = 1.4$ |
| (d) | M1 | For setting $3^x - 2 = Ax + B$ and obtaining $x = \log_3(Ax + (B+2))$ |
| | M1 | For equating coefficients to deduce that the equation of the line is $y = 6 - 3x$ |
| | A1 | For the correct equation of the line.<br>Minimally acceptable equation is $y = k - 3x$ where $k$ is a positive value |
| | ALT M1 | For rearranging the given equation |
| | ALT M1 | For taking exponents and rearranging. |
| | ALT A1 | For the correct equation of the line. |
| | M1 | For drawing their line on the graph.<br>The correct line will cross the axes at (0, 6) and (2, 0) |
| | A1 | For a value of 1.3 **only**<br>The calculator value is 1.291 |

## Type MS · Solving log and exponential equations with your graph

### Answer 7 of 12 · Question 8 part d · 5 marks

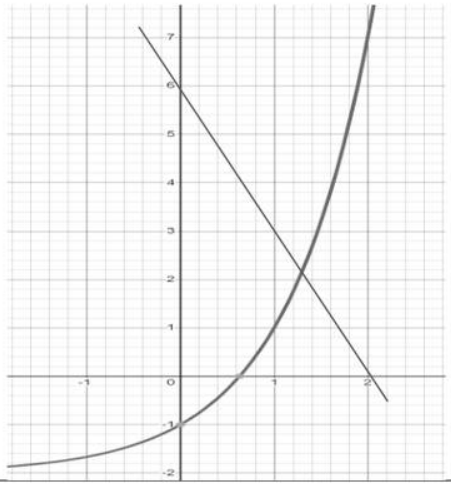
2025 NOV · Paper 01 · extract 3 of 3 · find part d in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|

Type MS · Solving log and exponential equations with your graph

Answer 8 of 12 · Question 8 part c · 3 marks

2025 NOV · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

| | | |
|---|---|---|
| **8(a)** | <table><tr><td>$x$</td><td>−1</td><td>−0.5</td><td>0</td><td>0.25</td><td>0.5</td><td>0.75</td><td>1</td><td>1.5</td><td>2</td></tr><tr><td>$y$</td><td>−1.7</td><td>−1.4</td><td>−1</td><td>−0.7</td><td>−0.3</td><td>0.3</td><td>1</td><td>3.2</td><td>7</td></tr></table> | B2<br>[2] |
| **(b)** |  | B2<br>[2] |
| **(c)** | $\log_3 4.5$<br>$y = 3^x - 2 \Rightarrow 3^x = y + 2 \Rightarrow x = \log_3(y+2)$<br>$\log_3 4.5 = \log_3(2.5+2) \Rightarrow y = 2.5$<br>$\Rightarrow x = 1.4$ | M1<br>M1<br>A1<br>[3] |
| **(d)** | $3^x - 2 = Ax + B \Rightarrow 3^x = Ax + (B+2) \Rightarrow x = \log_3(Ax + (B+2))$<br>$\Rightarrow \log_3(Ax + (B+2)) - x = 0$<br>$\log_3(8-3x) - x = 0$<br>$\Rightarrow A = -3 \quad 8 = B + 2 \Rightarrow B = 6$<br>$\Rightarrow y = 6 - 3x$<br><br>**ALT**<br><br>$\log_3(8-3x) - x = 0 \Rightarrow \log_3(8-3x) = x$<br>Taking exponents<br>$8 - 3x = 3^x \Rightarrow 8 - 3x - 2 = 3^x - 2 = y$<br>$6 - 3x = y$<br><br>Draws the line $y = 6 - 3x$<br>$x = 1.3$ | M1<br><br>M1<br><br>A1<br><br>[M1]<br><br>[M1]<br>[A1]<br><br>M1<br>A1<br>[5] |
| | | **Total 12 marks** |

## Type MS · Solving log and exponential equations with your graph

### Answer 8 of 12 · Question 8 part c · 3 marks

2025 NOV · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

| Part | Mark | Notes |
|---|---|---|
| 8 (a) | B2 | B2 for all correct values<br>B1 for a minimum of two values |
| (b) | B1ft | All points plotted correctly within half of one square. |
| | B1ft | Their points joined by a smooth curve. |
| (c) | M1 | **In this part, there must be evidence of algebraic work, together with use of the graph.**<br>For using log laws to rearrange the given expression to find $x = \log_3(y+2)$ |
| | M1 | For deducing that the required value of $y$ is 2.5 and using their graph to find the value of $x$ at that point. |
| | A1 | For $x = 1.4$ |
| (d) | M1 | For setting $3^x - 2 = Ax + B$ and obtaining $x = \log_3(Ax + (B+2))$ |
| | M1 | For equating coefficients to deduce that the equation of the line is $y = 6 - 3x$ |
| | A1 | For the correct equation of the line.<br>Minimally acceptable equation is $y = k - 3x$ where $k$ is a positive value |
| | ALT M1 | For rearranging the given equation |
| | ALT M1 | For taking exponents and rearranging. |
| | ALT A1 | For the correct equation of the line. |
| | M1 | For drawing their line on the graph.<br>The correct line will cross the axes at (0, 6) and (2, 0) |
| | A1 | For a value of 1.3 **only**<br>The calculator value is 1.291 |

## Type MS · Solving log and exponential equations with your graph

### Answer 8 of 12 · Question 8 part c · 3 marks

2025 NOV · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|

## Type MS · Solving log and exponential equations with your graph

### Answer 9 of 12 · Question 7 part c · 5 marks

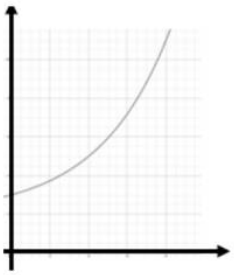
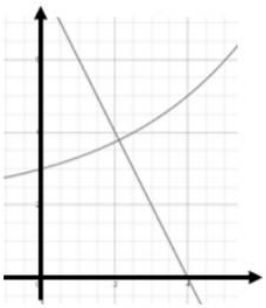
2021 JUN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

| | | |
|---|---|---|
| 7 (a) | <table><tr><td>2</td><td>3</td><td>4</td></tr><tr><td>3.73</td><td>4.28</td><td>5</td></tr></table> | B1 B1<br>(2) |
| (b) | Points plotted<br>Joined up with a smooth curve | B1ft<br>B1ft<br>(2) |
| (c) | $\log_3(6-2x)=\frac{x}{4}$ | M1 |
| | $6-2x=3^{\frac{x}{4}}$ | M1 |
| | $8-2x=3^{\frac{x}{4}}+2$ | A1 |
| | $y=8-2x$ drawn | M1 |
| | $x=2.1$ | A1<br>(5) |
| | | **Total 9 marks** |

Type MS · Solving log and exponential equations with your graph

Answer 9 of 12 · Question 7 part c · 5 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

<table>
<tr><th>Part</th><th>Mark</th><th>Guidance</th></tr>
<tr><td rowspan="2">(a)</td><td>B1</td><td>For two points (rounded correctly) correct from;<br><table><tr><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>3</td><td>3.32</td><td>3.73</td><td>4.28</td><td>5</td><td>5.95</td></tr></table></td></tr>
<tr><td>B1</td><td>All three points correct and correctly rounded.<br>Penalise rounding only once here. Condone 5.00</td></tr>
<tr><td rowspan="2">(b)</td><td>B1ft</td><td>All points plotted within half of one square. Ft their values of $y$ for $x = 2,3,4$ respectively</td></tr>
<tr><td>B1ft</td><td>All drawn points joined up in a smooth curve<br>  </td></tr>
<tr><td rowspan="5">(c)</td><td>M1</td><td>For use of power law to obtain $\log_3(6-2x) = \frac{\pm x}{4}$</td></tr>
<tr><td>M1</td><td>For removing the $\log_3$ to obtain: $6-2x = 3^{\frac{\pm x}{4}}$<br>Allow $(6-2x)^4 = 3^{\pm x}$ for this mark.</td></tr>
<tr><td>A1</td><td>For obtaining the equation $8-2x = 3^{\frac{x}{4}} + 2$ oe (eg., $-2x+8 = 2+3^{\frac{\pi}{4}}$)</td></tr>
<tr><td>M1</td><td>For drawing their straight line, provided it is of the form $y = k-2x$ where $k$ is a constant and $k \neq 6$<br>  <br>[Check coordinates (1, 6) (2, 4) (3, 2) (4, 0)]</td></tr>
<tr><td>A1</td><td>For the intersection point $(x =)$ 2.1</td></tr>
</table>

## Type MS · Solving log and exponential equations with your graph

### Answer 9 of 12 · Question 7 part c · 5 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

Type MS · Solving log and exponential equations with your graph

Answer 10 of 12 · Question 6 part d · 4 marks

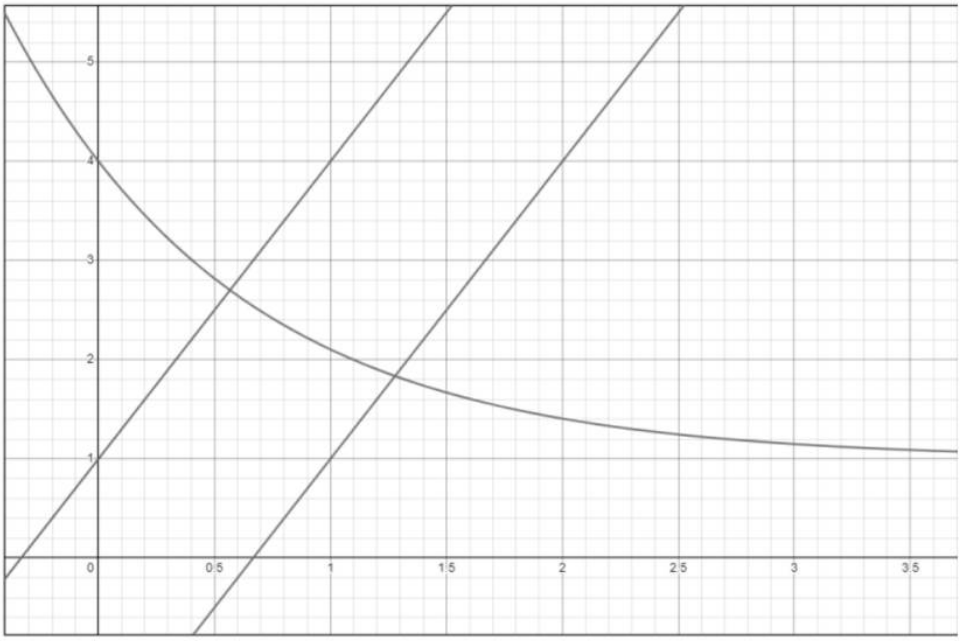
2022 JUN · Paper 01 · extract 1 of 3 · find part d in this mark scheme extract

| | | |
|---|---|---|
| 6 a | **0** \| *0.25* \| *0.5* \| **1** \| *1.5* \| **2** \| *3*<br>**4(.00)** \| *3.34* \| *2.82* \| **2.1(0)** \| *1.67* \| **1.41** \| *1.15* | B2<br>(2) |
| b | Points plotted within half a square<br>Points joined with a smooth curve | B1 ft<br>B1 ft<br>(2) |
| c | $x = e^{-x} \Rightarrow 3x + 1 = 1 + 3e^{-x}$ or sight of $y = 3x + 1$ and $y = 1 + 3e^{-x}$<br>$y = 3x + 1$ drawn. Intersection is at $x = 0.5$ or 0.6 | M1<br>M1 A1<br>(3) |
| d | $\ln(x-1)^3 = -3x \Rightarrow \ln(x-1) = -x$<br>$\Rightarrow 3x - 3 = 3e^{-x} \Rightarrow 3x - 2 = 1 + 3e^{-x}$<br>$y = 3x - 2$ drawn. Intersection is at $x = 1.2$ or 1.3 | M1<br>M1<br>M1 A1<br>(4) |
| ALT – first 2 marks | $(x-1)^3 = e^{-3x} \Rightarrow \left(\sqrt[3]{(x-1)^3} = \sqrt[3]{e^{-3x}}\right) \Rightarrow x - 1 = e^{-x}$<br>$3x - 3 = 3e^{-x} \Rightarrow 3x - 2 = 1 + 3e^{-x}$ | M1<br>M1 |
| | | **Total 11 marks** |

## Type MS · Solving log and exponential equations with your graph

### Answer 10 of 12 · Question 6 part d · 4 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part d in this mark scheme extract



## Type MS · Solving log and exponential equations with your graph

### Answer 10 of 12 · Question 6 part d · 4 marks

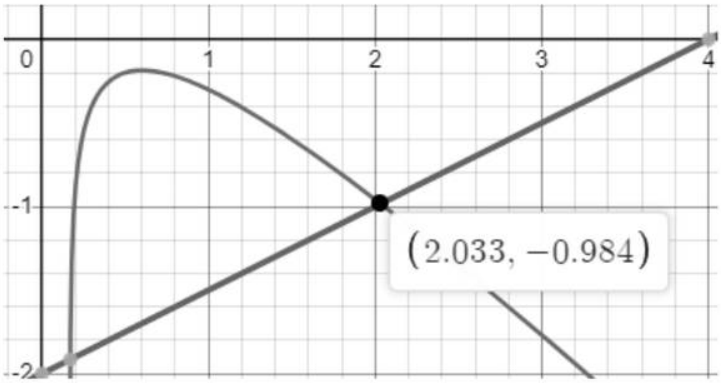
2022 JUN · Paper 01 · extract 3 of 3 · find part d in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|
| | $f(x) \ldots 4 \ldots 12 \ldots 19 \ldots$ | |

Type MS · Solving log and exponential equations with your graph

Answer 11 of 12 · Question 4 part c · 4 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

| | | |
|---|---|---|
| 4 (a) | $x$: 0.25, 0.5, 1, 1.5, 2, 2.5, 3<br>$y$: **-0.55**, -0.20, -0.30, -0.60, **-0.96**, **-1.35**, **-1.77** | B1<br>B1<br>[2] |
| (b) | Points plotted.<br>Smooth curve. | B1ft<br>B1ft<br>[2] |
| (c) | $\frac{3x-4}{2} = \log_{10}(6x-1)$<br>$\frac{x}{2} - 2 = \log_{10}(6x-1) - x$<br>Draws the line $y = \frac{x}{2} - 2$<br>2.0 / 2.1<br>  | M1<br>M1<br>M1<br>A1<br>[4] |
| | | **Total 8 marks** |

## Type MS · Solving log and exponential equations with your graph

### Answer 11 of 12 · Question 4 part c · 4 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

| Part | Mark | Additional Guidance |
|---|---|---|
| (a) | B1 | For 2 values correct. 2dp only. |
| | B1 | For all 4 values correct. 2dp only. |
| (b) | B1ft | For all 7 points plotted within half a square ft their values from the table. |
| | B1ft | For points joined with a smooth curve ft all their 7 points plotted, their points do not have be plotted correctly. |
| (c) | M1 | Correctly takes logs to give the expression shown.<br>Accept $\frac{3x-4}{2} = \log(6x-1)$ |
| | M1 | Rearranges to give an equation of the form<br>$\pm kx - 2 = \log_{10}(6x-1) - x$ (their left-hand side can be unsimplified equivalent) |
| | M1 | Draws a line of the form $y = \pm kx - 2$ oe, correctly. |
| | A1 | Accept answer given as 2.0 or 2.1 |
| ALT (c) | M1 | Correctly rearranges $y = \log_{10}(6x-1) - x$ to $10^{y+x} = 6x-1$ |
| | M1 | Equates $y + x$ to $\frac{3x-4}{2}$, and obtains $y = \pm kx - 2$ (can be unsimplified) |
| | M1 | Draws a line of the form $y = \pm kx - 2$ oe, correctly. There must be at least one intersection point with the curve. |
| | A1 | Accept answer given only as either 2.0 or 2.1, Do not allow 2 |

## Type MS · Solving log and exponential equations with your graph

### Answer 11 of 12 · Question 4 part c · 4 marks

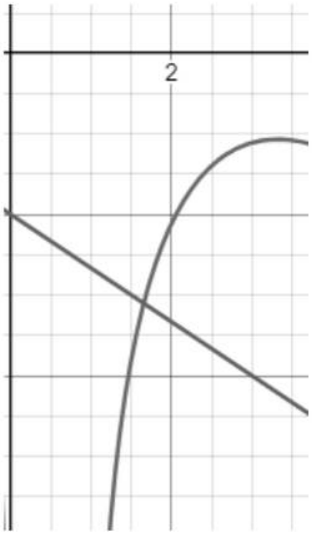
2024 NOV · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

Type MS · Solving log and exponential equations with your graph

Answer 12 of 12 · Question 7 part c · 6 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 4 · find part c in this mark scheme extract

| | | |
|---|---|---|
| 7 (a) | <table><tr><td>$x$</td><td>1.3</td><td>1.5</td><td>2</td><td>2.5</td><td>3</td><td>3.5</td></tr><tr><td>$y$</td><td>**−5.17**</td><td>−3.79</td><td>−2.11</td><td>**−1.39**</td><td>**−1.11**</td><td>**−1.08**</td></tr></table> | B1 B1<br><br>[2] |
| (b) | Points plotted.<br><br>Smooth curve.<br><br>  <br> | B1ft<br><br>B1ft<br>[2] |
| (c) | **General principle for marking**<br>M1 – Deal with $9 = 3^2$ or apply addition law and change $\log_3 9 = 2$<br>M1 – Deal with logs<br>M1 – Apply power law<br>A1 – Correct line<br><br>$\frac{3^{\frac{4}{3}x}}{3^2} = (x^2 - x)^3$<br>$3^{\frac{4}{3}x-2} = (x^2 - x)^3$<br>$\frac{4}{3}x - 2 = \log_3 (x^2 - x)^3$<br>$\frac{4}{3}x - 2 = 3\log_3 (x^2 - x)$<br>$-\frac{2}{3}x - 2 = 3\log_3 (x^2 - x) - 2x$ | <br><br><br><br>M1<br><br>M1<br><br>M1<br><br>A1 |

| | | |
|---|---|---|
| | **OR**<br>$3^{\frac{4}{3}x} = 9\left(x^2 - x\right)^3$<br>$\log_3 3^{\frac{4}{3}x} = \log_3 9\left(x^2 - x\right)^3 \Rightarrow \frac{4}{3}x = \log_3 9\left(x^2 - x\right)^3$<br>$\frac{4}{3}x = \log_3 9 + \log_3\left(x^2 - x\right)^3 \Rightarrow \frac{4}{3}x = 2 + \log_3\left(x^2 - x\right)^3$<br>$\frac{4}{3}x - 2 = \log_3\left(x^2 - x\right)^3 \Rightarrow \frac{4}{3}x - 2 = 3\log_3\left(x^2 - x\right)$<br>$-\frac{2}{3}x - 2 = 3\log_3\left(x^2 - x\right) - 2x$<br>Draws the line $y = \frac{-2}{3}x - 2$<br>$(x =)$ 1.6 or 1.7 | 2nd M1<br>1st M1<br>3rd M1<br>A1<br>M1<br>A1<br>[6] |
| ALT (c) | Set $3\log_3\left(x^2 - x\right) - 2x = Ax + B$<br>$\log_3\left(x^2 - x\right)^3 = Ax + 2x + B$<br>$\Rightarrow \left(x^2 - x\right)^3 = 3^{(A+2)x+B}$<br>Drawing on given equation<br>$9\left(x^2 - x\right)^3 = 9 \times 3^{(A+2)x+B} = 3^{(A+2)x+B+2}$<br>Equating exponents<br>$\frac{4}{3}x = (A+2)\text{x} \qquad 0 = B + 2$<br>$A = \frac{4}{3} - 2 = -\frac{2}{3} \qquad B = -2$<br>Hence $y = \frac{-2}{3}x - 2$<br>Draws the line $y = \frac{-2}{3}x - 2$<br>$(x =)$ 1.6 or 1.7 | M1<br>M1<br>M1<br>A1<br>M1<br>A1*cso<br>[6] |
| | | **Total 10 marks** |

## Type MS · Solving log and exponential equations with your graph

### Answer 12 of 12 · Question 7 part c · 6 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 4 · find part c in this mark scheme extract

| Part | Mark | Additional Guidance |
|---|---|---|
| 7 (a) | B1 | For 1 value correct to 2dp |
| | B1 | For all 4 values correct. |
| (b) | B1ft | For points plotted within half a square ft their values from the table. |
| | B1ft | For points joined with a smooth curve ft their points plotted. |
| (c) | **General principles using this method**<br>M1 – Deal with 9 or apply addition law and change $\log_3 9 = 2$<br>M1 – Deal with logs<br>M1 – Apply power law<br>A1 – Correct line | |
| | M1 | Correct rearrangement and use of index rule to give the equation shown.<br>[Deals correctly with $9 = 3^2$] |
| | M1 | Correctly takes logs with their expression to eliminate the exponential. |
| | M1 | Applies the power law for logs correctly with their expression |
| | A1 | For the equation seen oe |
| | M1 | Draws a line of the form $y = \frac{a}{b}x - 2$<br>Check the intercept on the $y$-axis or otherwise (0, –2) (1.5, –3) (3, –4) |
| | A1*cso | 1.6 or 1.7, 1 decimal place only. |
| ALT (c) | M1 | For equating the given formula for $y$ with a generic linear equation and setting out to find the coefficients and uses power law correctly. |
| | M1 | Correctly takes exponential from their expression to eliminate the.log |
| | M1 | Substitutes from their expression into the given equality. |
| | A1 | For finding the coefficients of the linear equation oe |
| | M1 | Draws a line of the form $y = \frac{a}{b}x - 2$<br>Check the intercept on the $y$-axis or otherwise (0, –2) (1.5, –3) (3, –4) |
| | ---------<br>A1*cso | 1.6 or 1.7, 1 decimal place only. |

## Type MS · Solving log and exponential equations with your graph

### Answer 12 of 12 · Question 7 part c · 6 marks

2025 JUN · Paper 01 (regional R) · extract 4 of 4 · find part c in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|